

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 28/05/2018

Day: Monday

Time: 10.00 a.m. to 10.30 a.m.

Total Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : MPT – May 2018
Semester: 1
Section : I
Course Code : PT758, PT764, PT770, PT788
Course Name: Research Methodology & Biostatistics

Important Instructions:

- There are two sections in the paper. **SECTION – I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION – I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes. Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION – II** immediately after handing over the **SECTION – I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

SECTION-I

Multiple Choice Questions: (MCQs)

(10 × 1 = 10)

(Encircle the correct answer)

1. Mean height of 10 female students of a class is 160 cm and the mean height of 20 male students is 172 cm. What will be the mean height of all the 30 students of the class?
 - a. 166 cm
 - b. 167 cm
 - c. 168 cm
 - d. 169 cm
2. The mean of 10 observations is 25 but later on it was found that an observation 24 was wrongly written as 14. What will be the mean of correct sample?
 - a. 24
 - b. 24.5
 - c. 26
 - d. 25

(P.T.O.)

3. A randomized trial comparing the efficacy of two drugs showed a difference between the two with a 'p' value of >0.05 . In reality, however, the two drugs do differ. This is therefore an example of:
 - a. Type I error
 - b. Type II error
 - c. Power of study
 - d. Sensitivity
4. A report shows the correlation coefficient calculated between variables height and weight in a study of 300 children of age 5-15 years from urban Anand as $r = 1.9$. What will be your interpretation?
 - a. Some calculation error
 - b. No correlation
 - c. Strong positive correlation
 - d. Moderate positive correlation
5. If the mean of a large approximately normally distributed sample is 230 and the standard deviation is 10, about 95% observations will be in this interval:
 - a. (210,250)
 - b. (220,240)
 - c. (225,235)
 - d. (230,210)
6. The best way to portray the most common symptoms in a newly detected diabetic patients is:
 - a. Box plot
 - b. Histogram
 - c. Pie chart
 - d. Frequency Polygon
7. The correlation coefficient between two variables (say X and Y) is 0.2. All the values of variable X are multiplied by 2. The correlation coefficient of the new data is:
 - a. -0.2
 - b. 0.2
 - c. 0.4
 - d. -0.4
8. You want to find association between gender (male/female) and prevalence of low back pain (yes/no). Which of the following graphical method you will prefer to do so?
 - a. Histogram
 - b. Box plot
 - c. Bar Chart
 - d. Frequency Polygon
9. All of the following are examples of probability sampling method EXCEPT :
 - a. Simple random sampling
 - b. Cluster sampling
 - c. Quota sampling
 - d. Two stage sampling
10. For a negatively skewed distribution:
 - a. Median is smaller than mean
 - b. Median is greater than mean
 - c. Tail to the right
 - d. Mean, Median and Mode are same

Identify the sentence as True or False:

(5 × 1 = 5)

(Please tick mark the answer)

True False

- | | | |
|---|--------------------------|--------------------------|
| 11. Correlation between two qualitative variables is graphically presented by scatter plot. | ☐ | ☐ |
| 12. The correct statement for a standard normal distribution is : Mean = 1, SD = 0. | ☐ | ☐ |
| 13. Median is the most popular measure of dispersion. | ☐ | ☐ |
| 14. Risk can be calculated from a cohort study. | ☐ | ☐ |
| 15. Choice of sampling technique can be decided irrespective of the study objective. | ☐ | ☐ |

Match the following

(5 × 1 = 5)

16.	Standard guidelines for reference writing	A. International Committee of Medical Journal Editors(ICMJE) B. Pedro C. Case Reports D. Descriptive study design E. Pilot testing F. Case fatality rate G. Incidence
17.	Study design to estimate the prevalence of a health problem in a community in 2016	
18.	Statistic used to estimate the risk or rate of acquiring a disease	
19.	A measure that reflects severity of an acute infectious disease	
20.	Study design for describing uncommon clinical manifestations	

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M.P.T. Examination

May 2018

PT758, PT764, PT770, PT788 Research Methodology & Biostatistics

Date: 28/05/2018

Time: 10.30 a.m. to 01.00 p.m.

Maximum Marks: 50

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

Q. 1. Short answers (5 out of 7) (Answer not exceeding 50 words) (5 × 2 = 10)

- a. Define type I and type II errors
- b. Enlist four characteristics of hypothesis.
- c. Advantages of case-control studies.
- d. Types of bias.
- e. What is qualitative research?
- f. Enlist components of research problem.
- g. Enlist measures of dispersion.

Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words) (4 × 5 = 20)

- a. Components of a research paper.
- b. Write short notes on the limitations and cautions in secondary data collection.
- c. Explain types of variables with example.
- d. Write a note on Internal and external validity.
- e. Advantage and disadvantage of non-probability sampling method.

Q. 3. Essay (2 out of 3) (2 × 10 = 20)

- a. What are the various methods of data collection? Discuss in detail along with their advantages and disadvantages?
- b. The procedure of testing hypothesis requires a researcher to adopt several steps. Describe in brief all such steps.
- c. Describe the different steps involved in a research process in detail.